6CCSACCM
Cooperation, Coordination and Multi-Agent Systems

Week 11

Voting (part II)

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Multiwinner elections

We distinguish the following three main types of multiwinner elections:

- Excellence-based elections
- Selecting a diverse committee
- Proportional representation

Challenges

Multiwinner elections lead to a number of challenges, of which we discuss the following two:

- Axiomatic properties
- Computational issues

Setting: ordinal vs. approval elections

- An election is a pair (C, R), where C is the set of candidates and R is the profile of voters' preferences.
- In the *ordinal* model, each voter i has a linear order $R_i = \succ_i \in \mathcal{L}(C)$ over all candidates, called their preference order or ordinal ballot.
- In the *approval* (dichotomous) model, each voter i has an approval set $A_i \subseteq C$ of candidates they approve, called their approval ballot.

Example: ordinal vs. approval elections

voter	ordinal ballot	approval ballot
1	$a \succ b \succ c \succ d \succ e$	$\{a, b, c\}$
2	$e \succ a \succ b \succ d \succ c$	$\{a, e\}$
3	$d \succ a \succ b \succ c \succ e$	$\{d\}$
4	$c \succ b \succ d \succ e \succ a$	$\{b, c, d\}$
5	$c \succ b \succ e \succ a \succ d$	$\{b, c\}$
6	$b \succ c \succ d \succ e \succ a$	$\{b\}$

Setting: multiwinner voting rule

- A *multiwinner voting rule* F takes an election (C, R) and a committee size k, $1 \leq k \leq |C|$, and returns a nonempty family of size-k subsets of C (the winning committees).
- In real applications, a separate tie-breaking scheme is typically used to pick a single winning committee from this family.

Single Transferable Vote (STV) rule

- We have an election with m candidates, n voters, and a target committee size k.
- STV runs in rounds and stops once k candidates have been elected.
- In each round:
  - Compute the quota $q = \lfloor \frac{n}{k+1} \rfloor + 1$.
  - If some candidate is ranked first by at least q voters, that candidate is elected to the committee.
  - Then, q voters who ranked this candidate first are removed from the election, and the candidate is deleted from all remaining preference orders.
  - If no candidate meets the quota q, the candidate with the fewest first-place votes is eliminated (removed from all ballots).

Example: STV for $k = 2$

voter	ordinal ballot
1	$a \succ b \succ c \succ d \succ e$
2	$e \succ a \succ b \succ d \succ c$
3	$d \succ a \succ b \succ c \succ e$
4	$c \succ b \succ d \succ e \succ a$
5	$c \succ b \succ e \succ a \succ d$
6	$b \succ c \succ d \succ e \succ a$

Families of multiwinner rules

- Committee scoring rules
- Rules for approval elections
- Rules based on the Condorcet principle

Committee scoring rules: setting

- Consider a setting with a set C of m candidates and with ordinal ballots.
- For a preference order $\succ$ and candidate c, we write $\mathrm{pos}_{\succ}(c)$ to denote the position of c in $\succ$.
- For a positive integer t, we write $[t]$ to denote the set $\{1, \ldots, t\}$.

Reminder: single-winner scoring functions

A *single-winner scoring function* $\gamma_m : [m] \to \mathbb{R}$, is a function that associates each position in a vote with a number of points, such that if $i < j$ then $\gamma_m(i) \geq \gamma_m(j)$.

- Borda:

$$\beta_m(i) = m - i$$

- t-approval:

$$\alpha_t(i) = \begin{cases} 1 & \text{if } i \leq t \\ 0 & \text{otherwise} \end{cases}$$

Reminder: single-winner scoring rules

A family $\gamma = (\gamma_m)_{m \in \mathbb{N}}$ of single-winner scoring functions defines a voting rule F_γ:

- For an election $E = (C, R)$ with $R = (\succ_1, \ldots, \succ_n)$, the score of a candidate c is

$$\mathrm{score}(c, E) = \sum_{i=1}^{n} \gamma_{|C|}(\mathrm{pos}_{\succ_i}(c))$$

- The rule elects the candidate(s) with the highest total score.

Committee positions

Let S be a size-k committee and $\succ$ a preference order.

- The position of S in $\succ$, denoted $\mathrm{pos}_{\succ}(S)$, is the list of the positions of the members of S, sorted in increasing order.
- $[m]_k$ denotes the set of all increasing sequences of length k chosen from $[m]$.
- For two committee positions $I = (i_1, \ldots, i_k)$ and $J = (j_1, \ldots, j_k)$ in $[m]_k$, I dominates J if $i_t \leq j_t$ for every t.

Committee scoring rules

A *committee scoring function* $\gamma_{m,k} : [m]_k \to \mathbb{R}$ for m candidates and committee size k, is a function that associates each committee position with a score in such a way that if $I, J \in [m]_k$ are two committee positions such that $I \succ J$, then $\gamma_{m,k}(I) \geq \gamma_{m,k}(J)$.

Definition

Let $\gamma = (\gamma_{m,k})_{k \leq m}$ be a family of committee scoring functions (one for each number m of candidates and committee size k). A *committee scoring rule* F_γ is a multiwinner rule that for election $E = (C, R)$, with $R = (\succ_1, ..., \succ_n)$, and committee size k outputs those committees W for which

$$\mathrm{score}(W, E) = \sum_{i=1}^{n} \gamma_{|C|,k}(\mathrm{pos}_{\succ_i}(W))$$

is the highest.

Committee scoring rules

Many well-known multiwinner rules are, in fact, committee scoring rules:

- **Single Non-Transferable Vote (SNTV).** Under SNTV, a committee receives a point from a voter if this committee contains the voters' most preferred candidate. That is, SNTV uses the scoring functions

$$\gamma_{m,k}^{\mathrm{SNTV}}(i_1, \ldots, i_k) = \alpha_1(i_1)$$

- **Bloc.** Under Bloc, each voter names his or her k favourite candidates and the winning committee consists of those mentioned most frequently. In other words, Bloc uses the scoring functions

$$\gamma_{m,k}^{\mathrm{Bloc}}(i_1, \ldots, i_k) = \sum_{t=1}^{k} \alpha_k(i_t)$$

Committee scoring rules

Many well-known multiwinner rules are, in fact, committee scoring rules:

- k-**Borda.** k-Borda outputs committee(s) that consist of k candidates with the highest (individual) Borda scores. That is, k-Borda uses the scoring functions

$$\gamma_{m,k}^{k-\mathrm{Borda}}(i_1, \ldots, i_k) = \sum_{t=1}^{k} \beta_m(i_t)$$

- **Chamberlin–Courant (β-CC).** The Chamberlin–Courant rule (β-CC) uses the scoring functions

$$\gamma_{m,k}^{\beta-\mathrm{CC}}(i_1, \ldots, i_k) = \beta_m(i_1)$$

  This means that the score that a committee receives from a voter is the Borda score of the committee member that the voter ranks highest (among all the committee members).

Example: Committee scoring rules for $k = 2$

voter	ordinal ballot
1	$a \succ b \succ c \succ d \succ e$
2	$e \succ a \succ b \succ d \succ c$
3	$d \succ a \succ b \succ c \succ e$
4	$c \succ b \succ d \succ e \succ a$
5	$c \succ b \succ e \succ a \succ d$
6	$b \succ c \succ d \succ e \succ a$

Condorcet committees

- **Fishburn, 1981:** A committee C is a *Condorcet committee* if for every other committee D (of the same size) a majority of voters prefers C to D.
  - Voters have explicit preferences over the committees, or there is an accepted mechanism for lifting preferences over candidates to those over committees.
- **Gehrlein, 1985; Ratliff, 2003:** Let (C, R) be an election, let k be a committee size, and let S be some committee of size k. We say that S is a *(weak) Condorcet set* if for every candidate c in S and every candidate d in $C \setminus S$ it holds that more than half (at least half) of the voters prefer c to d.

Stable multiwinner rules

A multiwinner rule is *stable* if it outputs a weak Condorcet set of a given size k whenever such a set exists.

- **Number of External Defeats (NED).** Under the NED rule, the score of a committee S is the number of pairs (c, d) of candidates such that $c \in S$, $dinC \setminus S$, and at least half of the voters prefer c to d. The committee(s) with the highest score are the winners.
- **Minimum Size of External Opposition (SEO).** Under the SEO rule, the score of a committee S is defined as

$$\min_{c \in S, d \in C \setminus S} |i \in N : c \succ_i d|$$

(i.e., the score of a committee S is the smallest number of voters that prefer some committee member to a committee nonmember). The committee(s) with the highest score are the winners.

Example: Condorcet stable rules for $k = 2$

voter	ordinal ballot
1	$a \succ b \succ c \succ d \succ e$
2	$e \succ a \succ b \succ d \succ c$
3	$d \succ a \succ b \succ c \succ e$
4	$c \succ b \succ d \succ e \succ a$
5	$c \succ b \succ e \succ a \succ d$
6	$b \succ c \succ d \succ e \succ a$

Approval-based rules

Let C be a set of m candidates, let k be the committee size, and let $w^{(k)} = (w_1^{(k)}, \ldots, w_k^{(k)})$ be a vector of k real numbers. The $w^{(k)}$-AV score that a voter with approval ballot A_i assigns to a committee S is $\sum_{j=1}^{|S \cap A_i|} w_j^{(k)}$.

Definition (Thiele, 1895)

Let $w = (w^{(i)})_{i \in \mathbb{N}}$ be a sequence of real-valued vectors (where each $w^{(i)}$ has i coordinates). Given an election (C, R) and a committee size k, the w-AV rule outputs those committees for which the sum of the $w^{(k)}$-AV scores assigned by the voters is the highest.

Thiele methods

- **Approval Voting (AV).** AV uses vectors $w^{(k)}$ of the form $(1, \ldots, 1)$. That is, AV outputs committees of those k candidates that are approved most frequently.
- **Approval-Based Chamberlin–Courant rule (α-CC).** Under the α-CC rule we use vectors of the form $(1, 0, \ldots, 0)$. As in the case of the ordinal-based Chamberlin–Courant rule (β-CC), a possible interpretation is that each voter chooses a representative from the committee and, thus, increases the score of the committee by one if there is at least one committee member that this voter approves.
- **Proportional Approval Voting (PAV).** The PAV rule uses vectors of the form $(1, 1/2, 1/3, \ldots, 1/k)$. This rule satisfies strong axioms pertaining to the proportionality of election results.

Example: Thiele methods for $k = 2$

voter	approval ballot
1	$\{a, b, c\}$
2	$\{a, e\}$
3	$\{d\}$
4	$\{b, c, d\}$
5	$\{b, c\}$
6	$\{b\}$

Proportionality

- In multiwinner elections, proportionality means the winning committee should reflect the support of different voter groups in rough proportion to their size.
- Intuitively, if a cohesive group of voters is large enough to "deserve" a seat, then the committee should include at least one candidate preferred by that group.
- Common formalisations of this idea include the axioms of Justified Representation (JR) and Extended Justified Representation (EJR), which capture the idea that sufficiently large groups of voters who agree on enough candidates should receive proportional influence over the final committee.

(E)JR

Definition

A rule satisfies *extended justified representation* (EJR) if for each approval election with n voters, each committee size k, and each $\ell \in [k]$, the following holds:
There is no group of $\lceil \ell \cdot n/k \rceil$ voters that all approve at least ℓ common candidates, but neither of whom approves ℓ or more members of each winning committee. A rule satisfies *justified representation* (JR) if it satisfies EJR for $\ell = 1$.

(E)JR

- AV fails JR and hence, EJR
- CC satisfies JR but fails EJR
- PAV satisfies EJR

THANK YOU!